

ML for Robots

Sensors:-

- i) Vision Sensors:- Cameras that capture and process image data for object detection.
- ii) Force Sensors:- devices that measure the force applied during interactions
- iii) Proximity Sensors:- Non-contact detectors that sense the presence of nearby objects using light.
- iv) Tilt Sensors:- Devices that measure the angles of inclination

Actuators:-

- i) Synchronous Motor:- AC motor where the rotor rotates at a constant speed synchronized with power supply frequency.
- ii) Stepper Motor:- digital motor that rotate with discrete steps for precise positioning without feedback.
- iii) AC Servo Motor:- A high precision AC motor paired with a controller for accurate closed loop position and speed control.
- iv) Brushless DC Servomotor:- A highly efficient, low-maintenance DC motor using electronic commutation for precise servo control.
- v) Brushed DC Servo Motor:- A traditional DC motor using brushes and a commutator; often paired with feedback for simple control.

→ (JOINTS)

1) Revolute (or Rotatory)

↳ Like a hinge

↳ Allows relative rotation about a fixed axis between two links

↳ Axis Rotation is the z -axis by convention.

2) Prismatic (or Linear)

↳ Like a piston

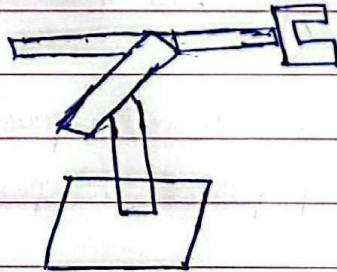
↳ Allows a relative translation along a fixed axis b/w two links.

i) Revolute and Prismatic joints are one degree of freedom (DOF) joints; thus they can be described using single numeric value called joint variable.

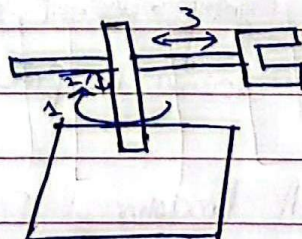
(i) $q_i = \theta_i$ (angle of rotation of link) (ii) $q_i = d_i$: displacement of link

↳ Revolute joints can have more than one degree of freedom (DOF)

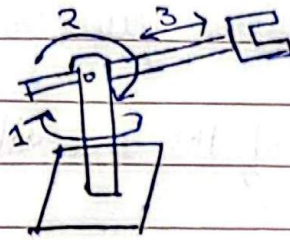
Cartesian: PPP



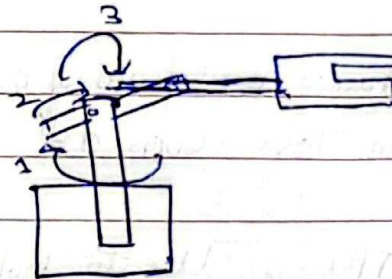
Cylindrical: RPP



Spherical :- RRP



Articulated :- RRR



> All revolute

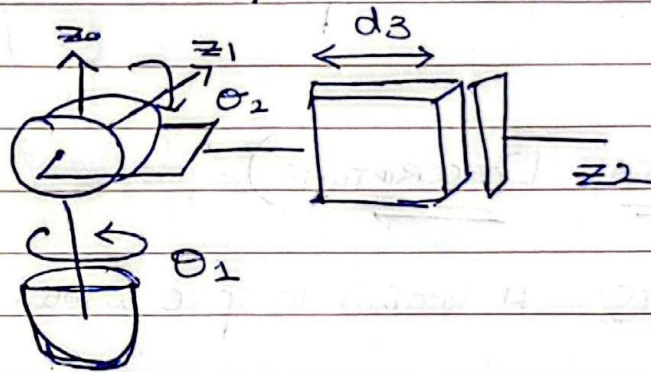
> z_0 : waist

> z_1 : shoulders (perpendicular to z_0)

> z_2 :- Elbow (parallel to z_1)

RRP :- Spherical :-

↳ Stanford Arm



SCARA :- RRP

EMBODIMENT / Work Envelop Concept :- 3D space within which a robot can manipulate its end-effector to perform tasks.

WORKSPACE :- set of all points in space that a robot's end-effector can reach, typically defined by its physical dimensions and joints limits.

↳ Wrist Configuration → Wrist assembly is attached to end of arm
→ end effector is attached to the wrist

WRIST DOF

> Pitch → Yaw → Roll

3 DOF wrist :- Roll, Pitch, yaw.

Arm $\rightarrow$ controls the position of the end effector & wrist controls the orientation.

End Effector :- device on the end of arm; attached via the wrist; that performs the task such as;

$\rightarrow$ Grippers :- Use to hold and move objects.

$\rightarrow$ Tools :- Used to perform work on a part; not just to pick it up.

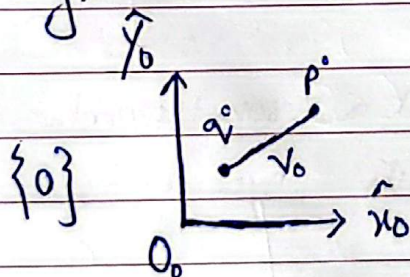
(SPATIAL DESCRIPTIONS)

POINTS :- A location in space without any direction.

Vector :- A quantity that has magnitude and direction.

$\rightarrow$ vectors and points use to describe the position and orientation of objects

e.g.:- end effector $\rightarrow$ point; force acting on robot arm $\rightarrow$ vector



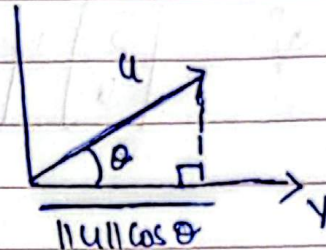
$$v_0 = P^0 - a_0 = \begin{bmatrix} 4 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

day / date:

→ DOT PRODUCT $\begin{cases} \text{Used to find projections of vectors} \\ \text{help in calculating angles b/w robots parts.} \end{cases}$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix}$$

$$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n = u^T v$$



$$1) \quad O_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \xrightarrow{\text{TRANSLATED}} O_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

Displacement vector from $\{0\}$ to $\{1\}$ is

$$d_1^0 = O_1 - O_0 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

↳ means; new frame shifted 3 units right from $\{0\}$.

$$\text{Given: } d_1^0 \begin{bmatrix} 3 \\ 0 \end{bmatrix} \{1\} P^1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

find P^0 in $\{0\}$;

$$\begin{aligned} \text{formula } P^0 &= d_1^0 + P^1 \\ &= \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \end{aligned}$$

P^1 shifted from $\{1\}$ to $\{0\}$; resulting new position (4,1) in $\{0\}$.

$$p^0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad q^0 = ? \quad d = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$q^0 = d + p^0$$

$$q^0 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(ROTATION)

Properties of Rotation Matrix

i) $R^T R = I$ (Orthogonality)

ii) Determinant = 1

$$\text{Det}(R) = 1$$

↳ Rotation preserves volume & shape.

↳ (Rotation in 2D)

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

↳ (ROTATION IN 3D)

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation

Point $\rightarrow P$ represented in polar form as

$$x = r \cos\phi \quad y = r \sin\phi$$

$\phi \rightarrow$ original angle before rotation $r \rightarrow$ distance from origin.

$$r = \sqrt{(r \cos\phi)^2 + (r \sin\phi)^2}$$

$$\theta = \tan^{-1} \left(\frac{r \sin\phi}{r \cos\phi} \right) \text{ or } \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

After rotating counterclockwise by θ ; the new angle become $(\theta + \phi)$

So new coordinate become $x' = r \cos(\theta + \phi)$ $y' = r \sin(\theta + \phi)$

Q) Given P in $\{0\}$ at $\begin{Bmatrix} 3 \\ 4 \end{Bmatrix} \begin{matrix} +x \\ +y \end{matrix}$ (1st Quadrant) Suppose

$\{0\}$ is rotated by an angle $\theta = 45^\circ$ we get $\{1\}$; determine the coordinate of new point P .

(i) $\theta = \tan^{-1}(y/x) = \tan^{-1}(4/3) = 53.13$ (ii) $P'_x = r \cos(\theta + \phi) = 5 \cos(45 + 53.13)$
 $\rightarrow P'_x = -0.7$
 $r = \sqrt{3^2 + 4^2} = 5$ $\rightarrow P'_y = r \sin(98.13) = 4.0$

Representation :- New point (p)

If $(x \text{ \& } y \text{ is } -ve)$ Then $180 - x$

→ Means in 3rd Quadrant $-y 270$

$$\theta = \tan^{-1}(y/x) \Rightarrow \boxed{\theta = 51} = (180 + 51) = \boxed{231}$$

↳ if $\Rightarrow (x \rightarrow +ve \text{ \& } y \rightarrow -ve) \rightarrow$ 4th Quadrant Then

$$\theta = \tan^{-1}(y/x) \Rightarrow \boxed{\theta = 20} \text{ suppose.}$$

$\Rightarrow$ then angle will be $(20 + 270) = \boxed{290}$

ROTATION MATRIX :-

P in $\{0\} \begin{Bmatrix} -4 \\ -5 \end{Bmatrix}$ suppose rotated by angle $\theta = 90^\circ$

$\{$ we get $\{1\}$; determine coordinate of new point "p"

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} -4 \\ -5 \end{bmatrix}$$

$$\theta = 90 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -4 \\ -5 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 \\ -4 \end{bmatrix}$$